Dietary Sodium Intake and Subsequent Risk of Cardiovascular Disease in Overweight Adults

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Context Dietary sodium is positively associated with blood pressure, and ecological and animal studies both have suggested that high dietary sodium intake increases stroke mortality.

Objective To examine the risk of cardiovascular disease associated with dietary sodium intake in overweight and nonoverweight persons.

Design Prospective cohort study.

Setting The first National Health and Nutrition Examination Survey Epidemiologic Follow-up Study, conducted in 1982-1984, 1986, 1987, and 1992.

Participants Of those aged 25 to 74 years when the survey was conducted in 1971-1975 (14 407 participants), a total of 2688 overweight and 6797 nonoverweight persons were included in the analysis.

Main Outcome Measures Dietary sodium and energy intake were estimated at baseline using a single 24-hour dietary recall method. Incidence and mortality data for cardiovascular disease were obtained from medical records and death certificates.

Results For overweight and nonoverweight persons, over an average of 19 years of follow-up, the total number of documented cases were as follows: 680 stroke events (210 fatal), 1727 coronary heart disease events (614 fatal), 895 cardiovascular disease deaths, and 2486 deaths from all causes. Among overweight persons with an average energy intake of 7452 kJ, a 100 mmol higher sodium intake was associated with a 32% increase (relative risk [RR], 1.32; 95% confidence interval [CI], 1.07-1.64; $P = .01$) in stroke incidence, 89% increase (RR, 1.89; 95% CI, 1.31-2.74; $P<.001$) in stroke mortality, 44% increase (RR, 1.44; 95% CI, 1.14-1.81; $P = .002$) in coronary heart disease mortality, 61% increase (RR, 1.61; 95% CI, 1.32-1.96; $P<.001$) in cardiovascular disease mortality, and 39% increase (RR, 1.39; 95% CI, 1.23-1.58; $P<.001$) in mortality from all causes. Dietary sodium intake was not significantly associated with cardiovascular disease risk in nonoverweight persons.

Conclusions Our analysis indicates that high sodium intake is strongly and independently associated with an increased risk of cardiovascular disease and all-cause mortality in overweight persons.

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Observational epidemiologic studies have repeatedly identified an independent, positive relationship between dietary intake of sodium and blood pressure across populations as well as within populations.[1-4] Randomized controlled trials have also demonstrated that reduced sodium intake leads to a reduction in blood pressure in both hypertensive and normotensive persons.[1,5-7] Given that blood pressure level is a strong risk factor for coronary heart disease and stroke, a high dietary sodium intake could be predicted to increase the risk of cardiovascular disease. Indeed, several ecological studies have found a positive relationship between average population dietary sodium intake and mortality due to stroke.[8-11] Animal studies have also indicated that a high-sodium diet increases stroke mortality in Dahl rats and stroke-prone spontaneously hypertensive rats.[12,13] However, several prospective cohort studies have failed to identify a significant association between dietary sodium intake and risk of stroke.[14,15] This may have been due to difficulties in measuring an individual's usual sodium intake or to use of relatively small sample sizes.[14,15]

In westernized populations, intraindividual variations in sodium intake are even greater than interindividual variations.[16] This measurement error diminishes the statistical power to detect significant associations between dietary sodium intake and cardiovascular outcomes in epidemiologic studies. Another difficulty in studying the relationship between sodium intake and cardiovascular disease is the heterogeneity of risk that exists at any given level of sodium intake because of the multifactorial nature of environmental and genetic influences on risk. For example, results of some epidemiologic and clinical studies suggest that obese persons are more sensitive to the effect of sodium on blood pressure than are their nonobese counterparts.[17-19] This enhanced sodium sensitivity may be due to in-

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creased renal tubular reabsorption of sodium in obese persons.[20] We made use of the large sample size and prolonged follow-up experiences of participants in the first National Health and Nutrition Examination Survey (NHANES I) Epidemiologic Follow-up Study to examine the risk of cardiovascular disease associated with dietary sodium intake in overweight persons.

METHODS

Study Population

In NHANES I, a multistage, stratified, probability-sampling design was used to select a representative sample of the US civilian noninstitutionalized population aged 1 to 74 years.[21,22] Certain population subgroups, including those with low incomes, women of childbearing age (25-44 years), and elderly persons (65 years or older) were oversampled. The NHANES I Epidemiologic Follow-up Study is a prospective cohort study of NHANES I participants who were 25 to 74 years of age when the survey was conducted in 1971 to 1975.[12-17] Of the 14 407 persons in this age range at baseline, we excluded 3059 who lacked 24-hour dietary recall information; 2 who lacked sodium intake information; 1133 who had a self-reported history of heart attack, heart failure, or stroke at baseline or had used medication for heart disease during the preceding 6 months; and 337 who were consuming a low-salt diet at baseline. Among remaining participants, 391 (4.0%) were lost to follow-up, leaving a total of 9485 participants, of whom 931 men and 1757 women were overweight (TABLE 1). Overweight was defined as a body mass index of 27.8 kg/m^2 or higher for men and 27.3 kg/m^2 or higher for women; nonoverweight was defined as a body mass index of less than 27.8 kg/m^2 for men and less than 27.3 kg/m^2 for women.[23]

Measurements

Baseline data collection included medical history, standardized medical examination, dietary history, laboratory tests, and anthropometric measurements.[21,22] A single 24-hour dietary recall was conducted by trained NHANES I personnel using a standardized protocol and 3-dimensional food-portion models. Frequency but not amount of salt added during food preparation or consumption was collected in the NHANES I. The dietary recall questionnaires were later coded by interviewers using nutrient information from the US Department of Agriculture Handbook No. 8[24] or other resources. Dietary sodium and energy intake were calculated for each participant by the National Center for Health Statistics. Frozen serum samples were sent to the Centers for Disease Control and Prevention for measurement of serum total cholesterol levels. Blood pressure, body weight, and height were obtained using standard protocols. The baseline questionnaire on medical history included questions about selected health conditions and medications used for those conditions during the preceding 6 months. Data on education, physical activity, and alcohol consumption were obtained by interviewer-administered questionnaires. Baseline information on smoking status was obtained in a random subsample of 936 overweight and 2313 nonoverweight participants who underwent more detailed baseline examination.[21,22] For the remaining study participants, information on smoking status at baseline was derived from responses to questions on lifetime smoking history at follow-up interviews in 1982 through 1984 or later.[25,26] Validity of information obtained using this approach has been documented.[25,26]

Follow-up Procedures

Follow-up data were collected between 1982 and 1984, and in 1986, 1987, and 1992.[27-30] Each follow-up examination included tracking a participant or his or her proxy to a current address; performing in-depth interviews with the participant or proxy; obtaining hospital and nursing home records, including pathology reports and electrocardiograms; and, for decedents, acquiring a death certificate. Incident cardiovascular disease was based on documentation of an event that met prespecified study criteria and occurred during the period between the participant's baseline examination and last follow-up interview. Mortality due to cardiovascular disease was based on death certificate reports. Validity of study outcome data from both sources has been documented.[31]

Incident stroke was based on death certificate reports in which the underlying cause of death was recorded using an *International Classification of Diseases, Ninth Revision* (*ICD-9*) code of 430-434.9, 436, or 437.0-437.1, or 1 or more hospital and/or nursing home stays in which the participant had a discharge diagnosis with one of these codes. Incident coronary heart disease was based on a death certificate report in which the underlying cause of death was coded as *ICD-9* 410-414, or by 1 or more hospital and/or nursing home stays in which the participant had a discharge diagnosis with an *ICD-9* code of 410-414. Cause-specific mortality was identified by underlying cause of death using the following *ICD-9* codes: 430-434.9, 436, or 437.0-437.1 (stroke); 410-414 (coronary heart disease); and 410-414, 430-434.9, 436, 437.0-437.1, 402-404, or 428 (cardiovascular disease). The date of record for incident events was identified by the date of first hospital admission with an established study event or date of death from a study event in the absence of hospital or nursing home documentation of such an event.

Statistical Analysis

Sodium intake and total energy intake were highly correlated in the study population ($r = 0.65$, $P < .001$). Therefore, both absolute sodium intake (mmol/d) and sodium-to-energy ratio (mmol/kJ) were used to examine the relationship between sodium intake and cardiovascular disease risk. Relative risk estimates were similar for both indexes. Sodium-to-energy ratio was expressed as 1 mmol of sodium per 7452 kJ, the average energy intake in the study population.

Because a statistically significant interaction on cardiovascular disease outcomes was detected between dietary sodium intake and overweight, all analyses were stratified by overweight. The quartile of sodium-to-energy ratio was cal-

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culated using the total sample (nonoverweight and overweight). For each baseline characteristic, mean value or corresponding percentage of study participants was calculated by quartile of sodium-to-energy ratio. The statistical significance of differences was examined by analysis of variance (continuous variables) and by the χ^2 test (categorical variables). The cumulative incidence of cardiovascular disease and mortality due to cardiovascular disease by quartile of sodium intake was calculated using the Kaplan-Meier method[32] and differences in cumulative rates were examined us-

Table 1. Baseline Characteristics According to Quartile of Dietary Sodium-to-Energy Ratio*

Variable	Quartile of Dietary Sodium-to-Energy Ratio (mmol/7452 kJ)				*P* Value for Trend
	0.8-62.3	>62.3-84.0	>84.0-112.3	>112.3-467.9	
2755 Nonoverweight Men and 4042 Nonoverweight Women					
No. of participants	1658	1725	1746	1668	. . .
Age, y	46.2 (15.4)	48.3 (15.8)	49.3 (15.9)	48.6 (15.8)	<.001
Male sex, %	37.9	39.5	42.7	42.0	.004
White race, %	82.3	87.6	86.3	90.1	<.001
Systolic blood pressure, mm Hg	129.0 (23.2)	129.5 (21.6)	131.4 (22.9)	130.7 (23.2)	.004
Diastolic blood pressure, mm Hg	80.6 (12.7)	80.2 (11.5)	81.3 (12.1)	80.6 (12.2)	.38
Hypertension, %†	19.1	19.1	21.6	21.8	.02
Serum cholesterol level, mmol/L	5.6 (1.2)	5.6 (1.2)	5.6 (1.3)	5.6 (1.2)	.52
Hypercholesterolemia, %‡	27.9	27.5	28.6	27.3	.89
Diabetes, %	2.1	2.6	2.9	3.8	.004
Body mass index, kg/m^2	23.1 (2.6)	23.1 (2.7)	23.1 (2.7)	23.2 (2.7)	.36
Low recreational physical activity, %	42.8	40.9	44.1	42.9	.53
Less than high school education, %	39.1	39.1	46.8	44.4	<.001
Current cigarette smoking, %§	43.7	36.7	34.9	38.3	.001
Daily alcohol drinking, %	31.6	25.8	22.8	23.6	<.001
Always use table salt, %	36.2	33.3	33.8	34.0	.25
Diuretic use, %	3.6	3.7	4.8	4.3	.14
Dietary intake of sodium, mmol/d	50.5 (27.0)	76.9 (34.6)	99.1 (44.5)	142.5 (69.3)	<.001
Dietary intake of energy, kJ/d	7983 (3946)	7828 (3506)	7636 (3347)	7243 (3376)	<.001
Sodium-to-energy ratio, mmol/7452 kJ	47.3 (11.9)	73.4 (6.1)	96.8 (8.1)	149.8 (40.6)	<.001
931 Overweight Men and 1757 Overweight Women					
No. of participants	713	647	625	703	. . .
Age, y	50.0 (14.9)	51.1 (15.0)	52.0 (15.0)	51.3 (14.8)	.07
Male sex	35.9	33.5	36.5	32.7	.41
White race	73.5	76.7	77.4	82.4	<.001
Systolic blood pressure, mm Hg	141.7 (24.1)	142.4 (24.4)	144.8 (25.2)	143.5 (24.6)	.07
Diastolic blood pressure, mm Hg	89.0 (12.9)	88.3 (13.0)	89.1 (13.4)	88.9 (13.2)	.77
Hypertension†	42.2	42.8	43.8	42.3	.89
Serum cholesterol level, mmol/L	5.9 (1.3)	5.9 (1.2)	6.0 (1.4)	5.9 (1.3)	.68
Hypercholesterolemia‡	39.1	38.6	38.4	36.4	.31
Diabetes	4.2	5.4	5.6	5.7	.21
Diuretic use	10.6	9.7	10.0	9.5	.55
Body mass index, kg/m^2	32.0 (4.4)	31.6 (4.1)	32.0 (4.7)	31.6 (3.9)	.13
Low recreational physical activity	52.1	57.7	51.9	54.9	.75
Less than high school education	54.5	56.4	57.4	56.8	.35
Current cigarette smoking‖	28.1	28.1	27.8	27.3	.71
Daily alcohol drinking	23.9	19.5	16.2	18.7	.006
Always use table salt	26.7	26.1	26.7	22.9	.14
Dietary intake of sodium, mmol/d	45.5 (28.3)	69.4 (33.6)	88.7 (47.3)	129.7 (73.6)	<.001
Dietary intake of energy, kJ/d	7209 (3962)	7045 (3339)	6845 (3707)	6468 (3217)	<.001
Sodium-to-energy ratio, mmol/7452 kJ	46.3 (12.7)	73.4 (6.3)	97.1 (8.2)	151.9 (43.0)	<.001

*All data are mean (SD) or percentage of participants; ellipses indicate not applicable.
†Systolic at least 160 mm Hg and/or diastolic at least 95 mm Hg and/or use of antihypertensive medication.
‡Serum cholesterol level at least 6.2 mmol/L. To convert cholesterol levels to milligrams per deciliter, multiply by 1/0.02586.
§Available for 2629 men and 3902 women.
‖Available for 892 men and 1690 women.

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ing the log-rank test for trend.[33] Cox proportional hazard models were used to explore the relationship between dietary sodium intake and cardiovascular disease risk.[34] Age was used as the time scale for all time-to-event analyses.[35] With the exception of stroke mortality models, for which there were few events in younger cohorts, all Cox proportional hazard models were stratified by birth cohort using 10-year intervals to control for calendar period and cohort effects.[35] Sodium intake was assessed both as a categorical (quartile) and continuous variable. Methods to estimate variances that take into account sample clustering and stratification of the NHANES I sample were used in Cox proportional

Table 2. Relative Risk (95% CI) of Cardiovascular Disease and Total Mortality According to Quartile of Dietary Sodium-to-Energy Ratio*

Variable	Quartile of Dietary Sodium-to-Energy Ratio (mmol/7452 kJ)				*P* Value for Trend
	0.8-62.3†	>62.3-84.0	>84.0-112.3	>112.3-467.9	
Among 2755 Nonoverweight Men and 4042 Nonoverweight Women					
No. of participants	1658	1725	1746	1668	. . .
Person-years	28 250	28 853	28 666	27 698	. . .
Stroke incidence	95	116	110	109	
Adjusted for age, race, and sex	1.0	1.03 (0.80-1.33)	0.92 (0.72-1.19)	0.99 (0.78-1.26)	.70
Multivariate-adjusted	1.0	1.05 (0.80-1.38)	0.93 (0.70-1.22)	0.95 (0.75-1.22)	.47
Stroke mortality	31	28	27	37	
Adjusted for age, race, and sex	1.0	0.79 (0.50-1.24)	0.71 (0.44-1.15)	1.08 (0.69-1.70)	.86
Multivariate-adjusted	1.0	0.75 (0.46-1.23)	0.62 (0.38-1.01)	0.88 (0.55-1.43)	.43
Coronary heart disease incidence	215	302	289	274	
Adjusted for age, race, and sex	1.0	1.24 (1.05-1.46)	1.07 (0.91-1.26)	1.09 (0.92-1.29)	.69
Multivariate-adjusted	1.0	1.34 (1.15-1.57)	1.05 (0.89-1.24)	1.06 (0.90-1.25)	.77
Coronary heart disease mortality	70	111	109	110	
Adjusted for age, race, and sex	1.0	1.31 (0.97-1.76)	1.16 (0.89-1.51)	1.26 (0.95-1.67)	.20
Multivariate-adjusted	1.0	1.57 (1.20-2.07)	1.26 (0.96-1.67)	1.37 (1.03-1.83)	.12
Cardiovascular disease mortality	112	151	147	156	
Adjusted for age, race, and sex	1.0	1.13 (0.87-1.46)	1.00 (0.81-1.24)	1.16 (0.92-1.48)	.37
Multivariate-adjusted	1.0	1.28 (0.99-1.67)	1.04 (0.83-1.31)	1.18 (0.92-1.51)	.46
Mortality from all causes	363	423	462	428	
Adjusted for age, race, and sex	1.0	1.01 (0.88-1.16)	1.01 (0.90-1.12)	1.01 (0.88-1.17)	.87
Multivariate-adjusted	1.0	1.09 (0.95-1.25)	1.04 (0.92-1.18)	1.02 (0.88-1.19)	.92
Among 931 Overweight Men and 1757 Overweight Women					
No. of participants	713	647	625	703	. . .
Person-years	11 920	10 644	10 037	11 188	. . .
Stroke incidence	45	61	75	69	
Adjusted for age, race, and sex	1.0	1.47 (0.98-2.20)	1.87 (1.24-2.83)	1.72 (1.22-2.42)	.002
Multivariate-adjusted	1.0	1.28 (0.86-1.89)	1.64 (1.07-2.52)	1.51 (1.06-2.14)	.02
Stroke mortality	15	17	24	31	
Adjusted for age, race, and sex	1.0	1.22 (0.61-2.46)	1.75 (0.93-3.27)	2.33 (1.33-4.09)	.002
Multivariate-adjusted	1.0	1.01 (0.49-2.11)	1.40 (0.73-2.70)	2.18 (1.21-3.94)	.005
Coronary heart disease incidence	169	160	158	160	
Adjusted for age, race, and sex	1.0	1.01 (0.82-1.24)	1.05 (0.85-1.30)	0.98 (0.79-1.22)	.97
Multivariate-adjusted	1.0	0.96 (0.78-1.19)	1.00 (0.80-1.25)	0.97 (0.77-1.21)	.86
Coronary heart disease mortality	52	46	55	61	
Adjusted for age, race, and sex	1.0	0.94 (0.63-1.40)	1.16 (0.78-1.71)	1.29 (0.96-1.73)	.05
Multivariate-adjusted	1.0	0.97 (0.64-1.47)	1.24 (0.83-1.86)	1.41 (0.96-2.06)	.03
Cardiovascular disease mortality	74	70	85	100	
Adjusted for age, race, and sex	1.0	1.02 (0.70-1.50)	1.27 (0.88-1.83)	1.52 (1.11-2.07)	.003
Multivariate-adjusted	1.0	1.01 (0.68-1.49)	1.28 (0.89-1.85)	1.63 (1.16-2.27)	<.001
Mortality from all causes	189	184	205	232	
Adjusted for age, race, and sex	1.0	1.07 (0.87-1.31)	1.22 (0.96-1.56)	1.39 (1.15-1.68)	<.001
Multivariate-adjusted	1.0	1.01 (0.81-1.27)	1.24 (0.97-1.60)	1.43 (1.19-1.72)	<.001

*Multivariate-adjusted data are stratified by birth cohort and adjusted for age, sex, race, systolic blood pressure, serum cholesterol level, body mass index, history of diabetes, diuretic use, physical activity, level of education, regular alcohol consumption, current cigarette smoking, and total energy intake. CI indicates confidence interval. Ellipses indicate not applicable.

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hazard models.[35] Data from the small number of participants who had reached 85 years of age were censored. Sex differences in the relationships between sodium-to-energy ratio and cardiovascular disease and all-cause mortality were tested using interaction terms in Cox proportional hazard models. Because there were no significant differences, men and women were pooled in the main analysis. However, a subgroup analysis by sex was also conducted.

RESULTS

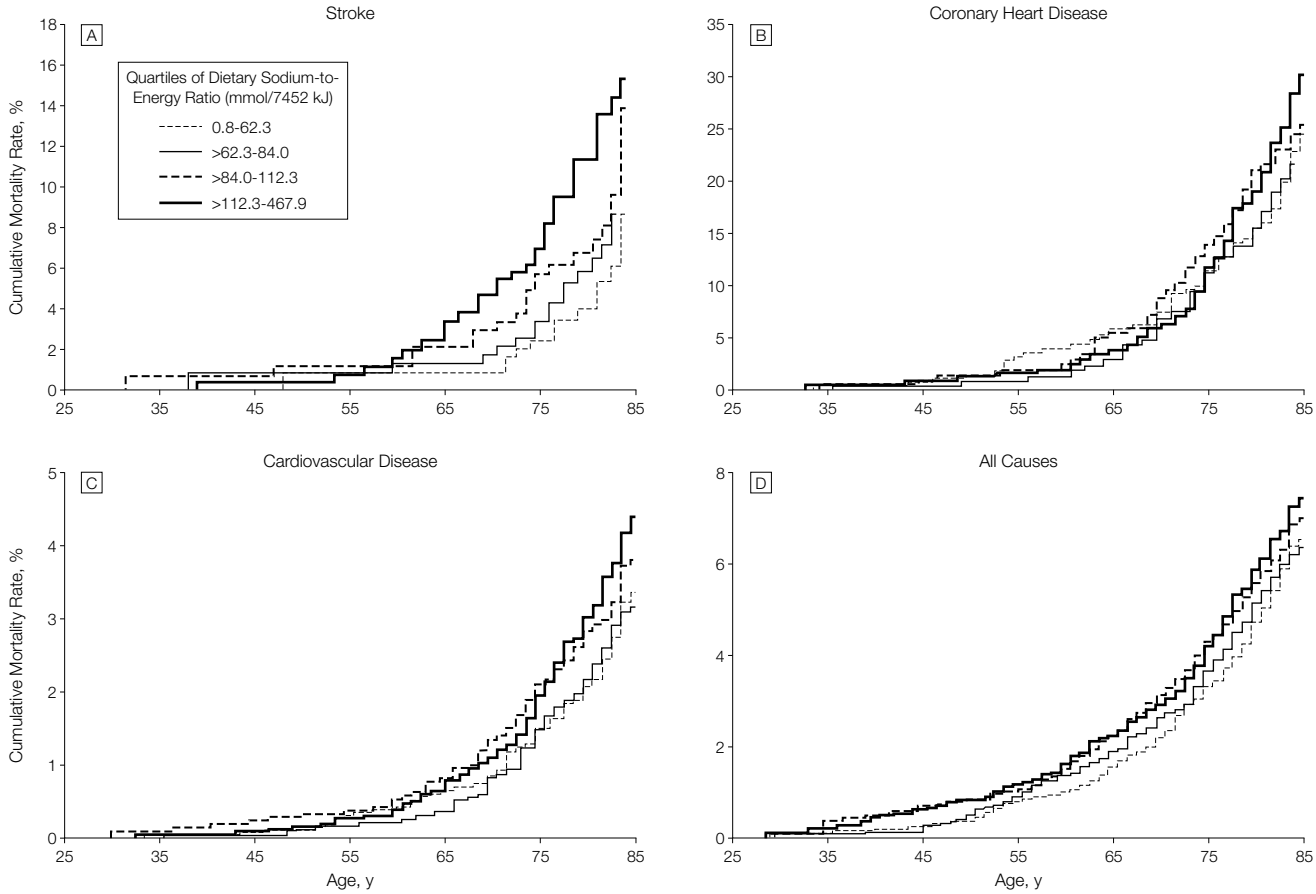
Compared with the study participants with a lower sodium intake, nonoverweight participants with a higher sodium intake were older; were more likely to be white and male; had higher mean systolic blood pressures; had higher prevalences of hypertension, diabetes, and low education; but had lower prevalences of current cigarette smoking and alcohol consumption (Table 1). In contrast, the means or percentages of most baseline variables were similar among the 4 sodium intake groups in the overweight participants except for the percentages of whites and regular alcohol drinkers.

During 113 467 person-years of follow-up from 1971 through 1992, 430 stroke events (123 fatal), 1080 coronary heart disease events (400 fatal), 566 cardiovascular disease deaths, and 1676 deaths from all causes were documented in the nonoverweight participants. Dietary sodium intake was not significantly associated with risk of cardiovascular disease in the nonoverweight persons (TABLE 2).

In age-, race-, and sex-adjusted analyses, stroke incidence and mortality, coronary heart disease mortality, and mortality due to cardiovascular disease and all causes were all positively and significantly associated with the corresponding trend in sodium-to-energy ratio. After additional adjustment for the factors listed in the first footnote to Table 2, all of the previously mentioned associations remained significant (Table 2). Similar results were obtained when quartile of sodium intake was used as the independent variable and when history of hypertension instead of systolic blood pressure was used as an adjustment variable.

During a total of 43 788 person-years of follow-up in the overweight participants, 250 stroke events (87 fatal),

Figure. Cumulative Mortality in 931 Overweight Men and 1757 Overweight Women



Data presented are cumulative mortality over an average follow-up of 19 years according to baseline dietary sodium to energy ratio from A, stroke (log-rank χ^2 for linear trend = 8.09, P = .004); B, coronary heart disease (log-rank χ^2 for linear trend = 1.91, P = .17); C, cardiovascular disease (log-rank χ^2 for linear trend = 7.11, P = .008); and D, all causes (log-rank χ^2 for linear trend = 8.87, P = .003).

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Table 3. Multivariate Relative Risk (95% CI) of Cardiovascular Disease and Total Mortality Associated With a 100-mmol Increase in Dietary Sodium Intake Among Nonoverweight and Overweight Participants*

	Nonoverweight (n = 6797)	Overweight (n = 2688)	*P* Value†
Sodium-to-Energy Ratio (100 mmol/7452 kJ)			
Stroke incidence	0.98 (0.83-1.16)	1.32 (1.07-1.64)	.03
Stroke mortality	0.90 (0.63-1.28)	1.89 (1.31-2.74)	<.001
Coronary heart disease incidence	0.95 (0.83-1.10)	1.06 (0.88-1.29)	.39
Coronary heart disease mortality	1.07 (0.87-1.31)	1.44 (1.14-1.81)	.07
Cardiovascular disease mortality	1.02 (0.85-1.22)	1.61 (1.32-1.96)	.003
Mortality from all causes	1.00 (0.90-1.11)	1.39 (1.23-1.58)	<.001
Dietary Sodium Intake (100 mmol/d)			
Stroke incidence	0.99 (0.81-1.21)	1.39 (1.09-1.77)	.02
Stroke mortality	0.82 (0.55-1.22)	1.98 (1.25-3.14)	.003
Coronary heart disease incidence	0.96 (0.86-1.08)	0.94 (0.76-1.17)	.87
Coronary heart disease mortality	1.07 (0.89-1.28)	1.29 (1.01-1.64)	.22
Cardiovascular disease mortality	1.00 (0.84-1.19)	1.45 (1.20-1.75)	.006
Mortality from all causes	0.98 (0.88-1.09)	1.32 (1.16-1.50)	.002

*Data are stratified by birth cohort and adjusted for age, sex, race, systolic blood pressure, serum cholesterol level, body mass index, history of diabetes, diuretic use, physical activity, level of education, regular alcohol consumption, current cigarette smoking, and total energy intake. CI indicates confidence interval.
†*P* value for interaction between sodium intake and body weight (nonoverweight vs overweight).

647 coronary heart disease events (214 fatal), 329 cardiovascular disease deaths, and 810 deaths from all causes were documented. The cumulative mortality of stroke at age 85 years was 9.0%, 8.9%, 14.4% and 15.8% among patients within the first, second, third, and fourth quartiles of sodium-to-energy ratio, respectively ($P = .004$ for trend) (FIGURE). The corresponding cumulative estimates by quartile of sodium-to-energy ratio were 24.9%, 22.3%, 26.0%, and 30.9% for coronary heart disease mortality ($P = .17$ for trend); 34.3%, 32.4%, 39.0%, and 44.9% for cardiovascular disease mortality ($P = .008$ for trend); and 65.8%, 63.7%, 70.2%, and 74.7% for mortality from all causes ($P = .003$ for trend), respectively. The cumulative incidence of stroke at age 85 years was 23.7%, 30.4%, 41.4%, and 33.9% among patients within the first, second, third, and fourth quartiles of sodium-to-energy ratio, respectively ($P = .005$ for trend). The cumulative incidence of coronary heart disease was not significantly associated with the corresponding sodium-to-energy ratio.

There were statistically significant interactions on major outcomes between sodium intake and overweight (TABLE 3). Dietary sodium intake was significantly associated with increased stroke incidence and mortality from stroke, coronary heart disease, cardiovascular disease, and all causes in overweight but not in nonoverweight persons. Furthermore, the RRs were similar when either sodium-to-energy ratio or absolute sodium intake was used as the independent variable.

The association between sodium intake and risk of cardiovascular disease was similar in men and women (*P* value for interaction varied from .19 to .98). For example, a 100-mmol per 7452 kJ higher intake of sodium was associated with a 67% increase (RR, 1.67; 95% CI, 1.27-2.18) in cardiovascular mortality in men and a 53% increase (RR, 1.53; 95% CI, 1.13-2.08) in women ($P = .69$ for interaction). A 100-mmol per 7452 kJ higher intake of sodium was associated with a 44% increase (RR, 1.44; 95% CI, 1.14-1.81) in all-cause mortality in men and a 35% increase (RR, 1.35; 95% CI, 1.13-1.62) in women ($P = .71$ for interaction).

COMMENT

Our study is the first to document the presence of a positive and independent relationship between dietary sodium intake and cardiovascular disease risk in adults. In 1995, an estimated 960 592 US residents died of cardiovascular disease, representing 41.5% of all deaths.[36] In many economically developing countries, cardiovascular disease mortality has increased rapidly and has become the leading cause of death.[37] Dietary sodium reduction has been recommended as a means to prevent both hypertension and cardiovascular disease mortality and morbidity.[38,39] Our findings suggest that reduced sodium intake may be especially efficacious in overweight persons.

Obesity activates the sympathetic nervous and renin-angiotensin systems, causes insulin resistance and hyperinsulinemia, and alters intrarenal vascular resistance. These changes have been related to enhanced renal tubular sodium reabsorption and sodium retention.[20,40] In a study of 60 obese and 18 nonobese adolescents, Rocchini et al[18] found that blood pressure was more readily affected by dietary sodium intake in obese than nonobese adolescents, and this increased sodium sensitivity was reduced after weight loss. Increased sodium sensitivity in overweight persons may explain why we identified a strong, independent relationship between dietary sodium intake and cardiovascular disease risk, whereas other investigators who have studied predominantly nonoverweight populations have not found such an association.[14,15] Overweight is a common and important cardiovascular disease risk factor. According to data from NHANES III, conducted from 1988 to 1991, 33.4% of US adults aged 20 years or older were overweight as defined by a body mass index of 27.8 kg/m^2 or higher for men and 27.3 kg/m^2 or higher for women.[41] That the prevalence of obesity has increased progressively during recent decades in males and females of all racial groups in the United States is particularly troubling.[41] To reduce the cardiovascular disease risk in overweight patients, both weight loss and sodium reduction should be recommended. For persons with difficulty losing weight, greater attention to reductions in sodium intake may be appropriate.

Animal studies have identified a significant increase in stroke mortality in

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Dahl salt-resistant rats that were placed on high-salt diets compared with those that were maintained on lower-salt diets, despite similar levels of blood pressure in the 2 groups.[42] Findings from ecological analyses indicating that the association between sodium intake and stroke risk was stronger than the corresponding relationship between sodium intake and blood pressure level may also imply a direct effect of sodium intake on stroke risk.[9,11] In our study, dietary sodium intake was related to cardiovascular disease risk and total mortality, independent of baseline levels of systolic blood pressure. Moreover, estimated RRs of cardiovascular disease associated with a 100-mmol greater sodium intake were much larger than would be expected based on the corresponding influence on blood pressure. In clinical trials, a 100-mmol reduction in dietary sodium intake was associated with an average reduction in diastolic blood pressure of 1.4 to 2.5 mm Hg.[5] This could potentially result in a 15% decrease in stroke risk.[43] Our study estimated that a 100-mmol difference in dietary sodium intake was associated with a 32% increase in stroke risk. The apparent excess of observed RR of cardiovascular disease associated with sodium intake compared with that expected on the basis of anticipated blood pressure increase suggests the possibility of an independent direct effect of sodium intake on cardiovascular disease.

Alderman and colleagues[44] used data from the NHANES I Epidemiologic Follow-up Study to examine the relationship between dietary sodium intake and mortality from cardiovascular disease and all causes. In contrast with our findings, they identified an inverse relationship between sodium intake and mortality from cardiovascular disease ($P = .09$) and all causes ($P < .007$) and a positive relationship between sodium-to-energy ratio and mortality from cardiovascular disease ($P = .006$) and all causes ($P < .001$). However, several methodological concerns make it difficult to interpret their findings. For instance, they did not exclude from their main analysis participants with baseline histories of cardiovascular disease, albeit such participants might have been expected to have changed their dietary intake of sodium. In addition, they did not exclude participants who were consuming low-sodium diets at baseline. Acute rheumatic fever, chronic rheumatic heart disease, and diseases of the pulmonary circulation were included as cardiovascular mortality outcomes, although there is no obvious biological basis for a relationship between sodium intake and these diseases. Perhaps of greatest concern is the fact that they included sodium intake, energy intake, and sodium-to-energy ratio as continuous variables in the same multivariate model. Given that an interaction term was included in their analysis model, it is not possible to interpret the main effect of sodium intake alone on the outcomes of interest. The inconsistency of the association between the 2 sodium intake indicators (sodium alone and sodium-to-energy ratio) and outcomes of interest in the study by Alderman et al might reflect the heterogeneity of this relationship in a study population with different body weights. In our analyses, we found a consistent relationship between both sodium intake and sodium-to-energy ratio with risk of cardiovascular disease and total mortality in overweight persons but not in nonoverweight persons.

In an earlier study, Alderman and colleagues[45] reported a significant inverse association between urinary sodium excretion and incidence of myocardial infarction in a prospective study of 2937 patients treated for hypertension. Unmeasured variables and imprecision of the potentially confounding variables that were measured might have contributed to the occurrence of this unexpected finding.[46,47] In addition, urinary sodium excretion was measured after 5 days of dietary sodium restriction, which most likely would not have provided a valid assessment of habitual intake of dietary sodium. Finally, the study was conducted in hypertensive patients who were enrolled in a work-site treatment program. As such, the findings may not have general applicability. In our study, dietary sodium intake was not associated with coronary heart disease incidence but with increased coronary heart disease mortality. Similarly, analyses from the Multiple Risk Factors Intervention Trial cohort did not find an inverse association between sodium intake and incidence of myocardial infarction.[48]

A limitation of our study is the estimation of sodium intake by a single 24-hour dietary recall. This may result in misclassification of usual sodium intake at the individual level. In addition, the dietary recall method used in NHANES I may have underestimated sodium intake because it did not include quantitative data on discretionary use of salt in the cooking or seasoning of food at the table, settings that account for about 15% to 30% of the sodium intake in Western societies.[49,50] Likewise, NHANES I study participants may have underreported their dietary energy intake.[51,52] These measurement errors would tend to bias our RR estimates toward 1 (effect to 0) in univariate models. Aside from a possible lowering of sensitivity to sodium intake in nonoverweight compared with overweight persons, these measurement errors in sodium intake might have contributed to our finding of a null association in nonoverweight persons. Another limitation is that participants were passively followed up for clinical outcomes. However, any resultant misclassification bias is likely to have been confined to characterization of incident cases and should have had no effect on estimation of mortality experience.

Our study has several important strengths. First, the findings can be generalized to overweight persons in the US general population because the NHANES I Epidemiologic Follow-up Study cohort is a random sample of this population. In addition, temporal relationships can be established with confidence because dietary sodium intake was measured at baseline, and subsequent cardiovascular disease and total mortality were assessed over an average of 19 years. The fact that follow-up experience was available for more than

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96% of study participants further enhances the validity of our findings.

Our study indicates that a high sodium intake is strongly and independently associated with an increased risk of cardiovascular disease in overweight persons. While we await additional studies on this important public health issue, our findings support existing recommendations for moderate reduction in sodium intake aimed at reducing blood pressure and cardiovascular disease risk in the overweight adult population.

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